GE-Problem Set 2, ECN 714

Exercise 1 Consider a one consumer private ownership economy with two goods in which

$$\begin{aligned} U(x,y) &= \log x + \log y \\ \omega &= (1,1) \\ y &= f(x) = (x+a)^{0.5} \\ a &> 0. \end{aligned}$$

a. Construct the production set for the firm in this economy.
b. Find the equilibria of this economy. Do not ignore corner solutions.

Exercise 2 Consider an exchange economy with two consumers, A and B, and three commodities, x, y and z. The utility functions and endowment vectors are:

$$\begin{aligned} U_A(x,y,z) &= xyz^2, w_A = (2,1,3) \\ U_B(x,y,z) &= xy^2z, w_B = (2,3,1) \end{aligned}$$

Please find the Walrasian equilibria (i.e., prices and allocations) for this economy.

Exercise 3 Consider the following optimization problem in an exchange economy with two agents, a and b, and two goods, 1 and 2:

$$\begin{aligned} &\max_{x_a\ ,\ x_b} U_a(x_a) \\ &s.t. \\ U_b(x_b) &\geq \bar{U}_b \\ x_a + x_b &\leq \omega_a + \omega_b \\ x_a, x_b &\geq 0 \end{aligned}$$

Each consumer's utility function is continuous and strongly monotone. Please prove that any feasible allocation which is Pareto Optimal must also be a solution to the described optimization problem for some value $\bar{U}_b$.

Exercise 4 A converse to the separating hyperplane theorem. Suppose that $B \subset \mathbb{R}^N$ is a closed set. Also suppose that for any $x \notin B$, $\exists p \in \mathbb{R}^N$ with $p \neq 0$ and a value $c \in \mathbb{R}$ such that $p \cdot x > c$, $p \cdot y < c$, $\forall y \in B$ (i.e. for any point x outside the set B, there is a hyperplane that separates B and x). Please prove that B is a convex set.

Exercise 5 Consider the exchange economies below. Each economy has two agents and two goods. The utility functions and initial endowments are as specified below. Calculate the Pareto set and the set of Walrasian equilibria and illustrate them in an Edgeworth box. Do not ignore possible boundary solutions.

a.
$$U_1(x,y) = \log x + \log y$$
$$U_2(x,y) = 0.5\log x + \log y$$
$$w_1 = w_2 = (0.5, 0.5)$$

b.
$$U_1(x,y) = x + y$$
$$U_2(x,y) = 2x + y$$
$$w_1 = w_2 = (0.5, 0.5)$$

c.
$$U_1(x,y) = \log x + y$$
$$U_2(x,y) = \log x + 2y$$
$$w_1 = (1,0), \quad w_2 = (0,1)$$